Weekly Review and Practical Report

Task 1: Configure a DNS server role on your Windows Server VM and create a new forward lookup zone named 'myapp.local' with an A record pointing to 192.168.1.50.

Steps performed

1. Installed the DNS Server role from Server Manager (Add Roles and Features).
2. In DNS Manager, right-clicked Forward Lookup Zones, chose New Zone, selected Primary zone and named it myapp.local.
3. Right-clicked the new zone, chose New Host (A or AAAA), entered the name www and the IP address 192.168.1.50.
4. The existing foodies.local zone was left untouched, so the earlier lab was not affected. Two zones can point to the same IP address without conflict.

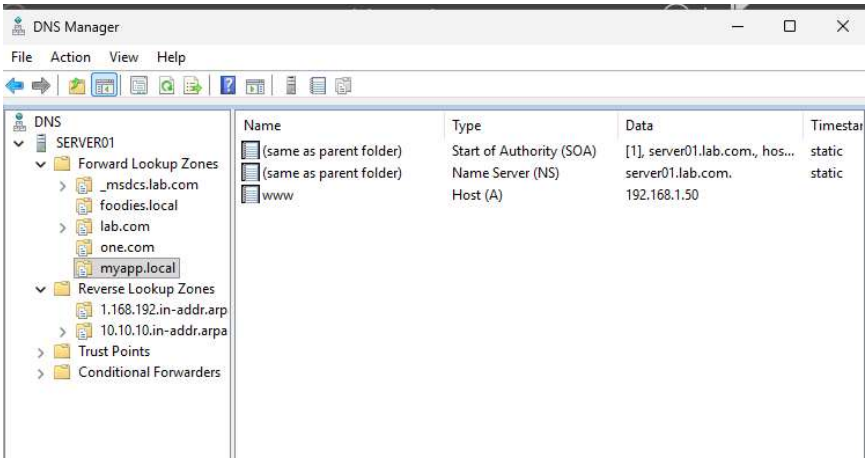


Figure 1: myapp.local zone with the www A record (192.168.1.50)

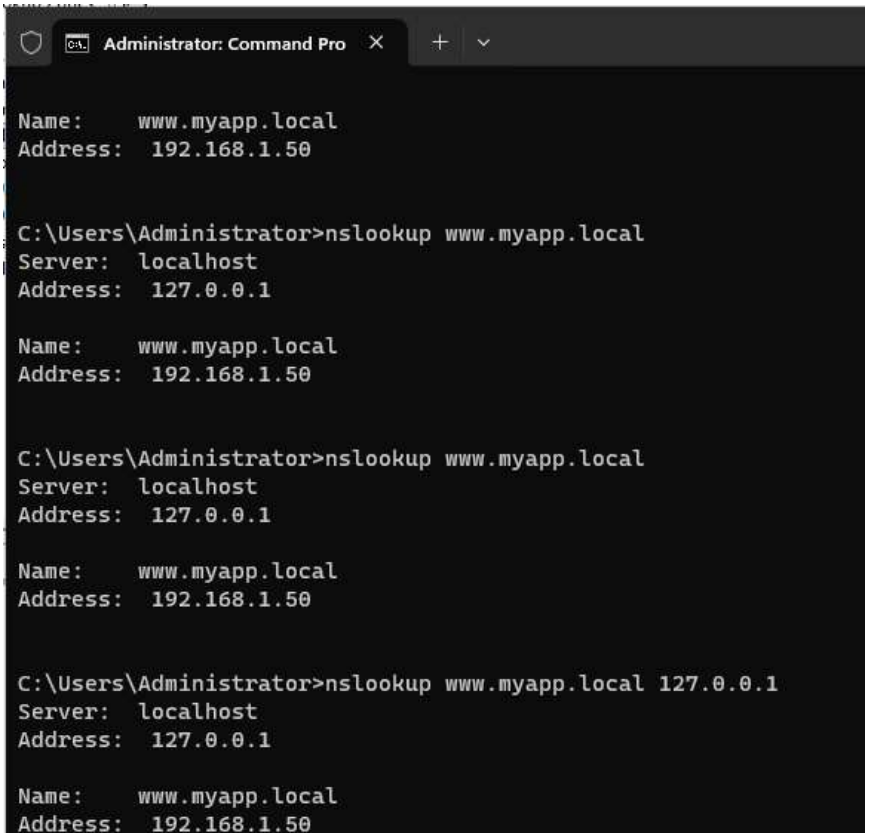


Figure 2: nslookup www.myapp.local returns 192.168.1.50

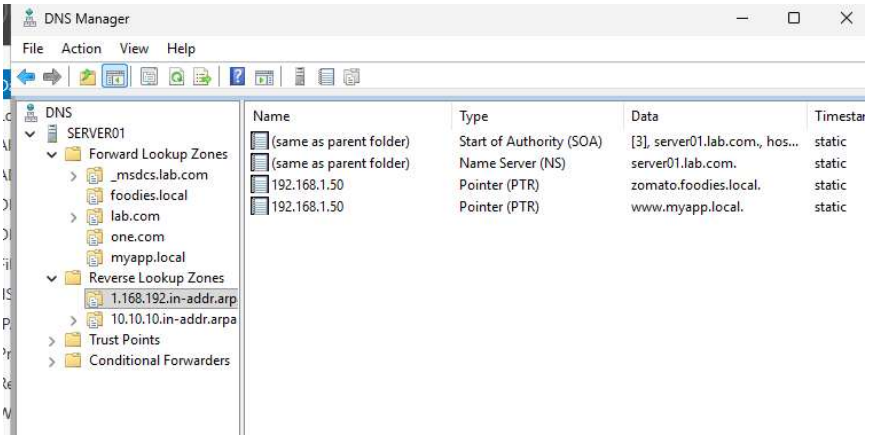


Figure 3: Reverse zone PTR records for 192.168.1.50

nslookup resolved www.myapp.local to 192.168.1.50 using the local DNS server (127.0.0.1), so the zone and record work.

Task 2: Set up the DHCP server role and create a new scope that assigns IP addresses from 192.168.1.100 to 192.168.1.120 with a lease duration of 8 hours.

Steps performed

1. Installed the DHCP Server role.
2. In the DHCP console, right-clicked IPv4 and chose New Scope.
3. Named the scope myapp.local, set the start address 192.168.1.100, the end address 192.168.1.120 and the subnet mask 255.255.255.0 (length 24).
4. Set the lease duration to 0 days, 8 hours, 0 minutes. The wizard first had 8 days, which was corrected in the scope Properties.

5. Activated the scope.

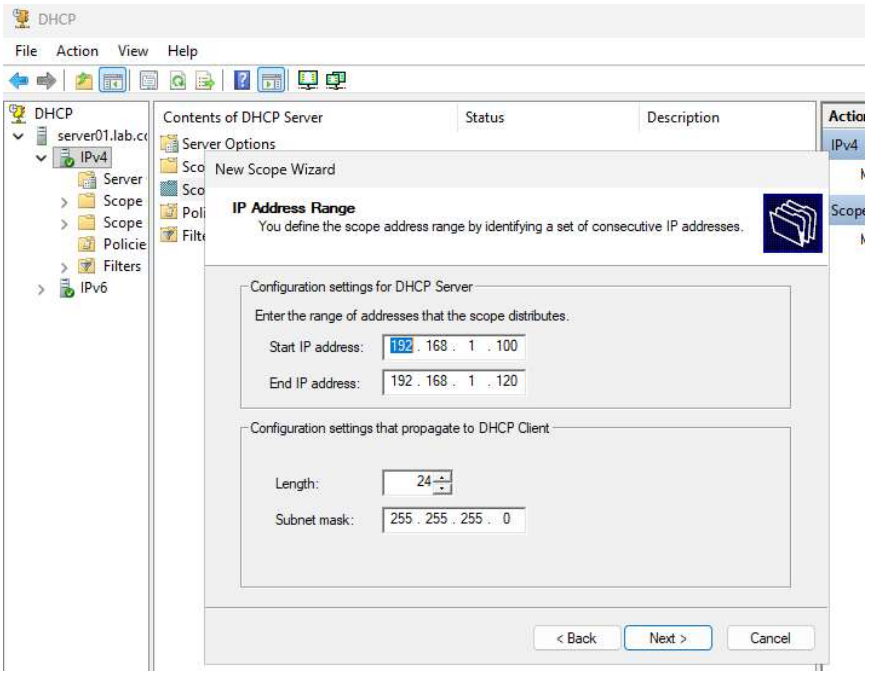


Figure 4: New Scope Wizard, address range 192.168.1.100 to 192.168.1.120

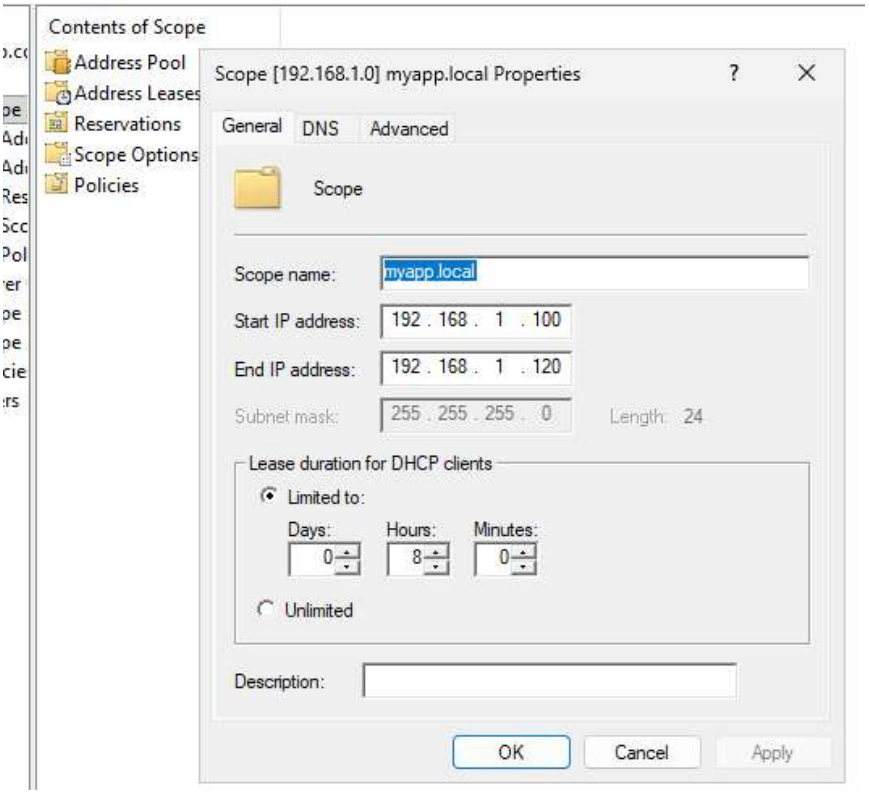


Figure 5: Scope Properties, lease duration set to 0 days, 8 hours

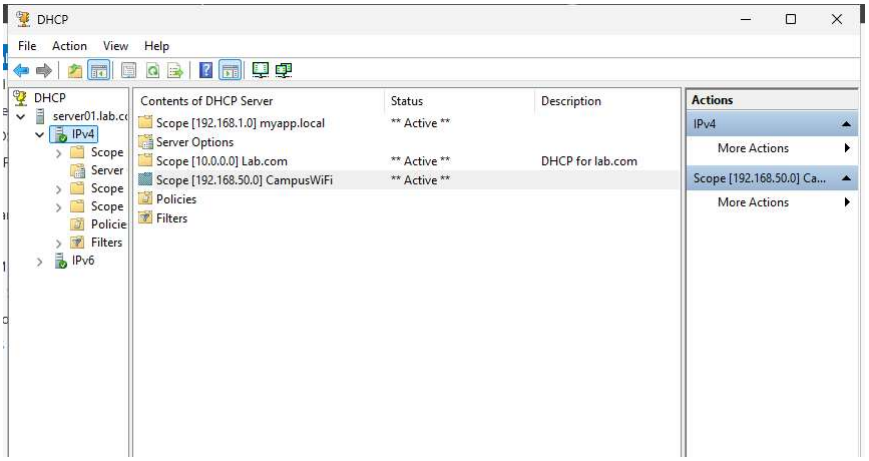


Figure 6: Scope [192.168.1.0] myapp.local shown as Active

Task 3: Share a folder named 'MusicLibrary' on your Windows Server and set permissions so only users in the 'Students' group can read and write to it.

Steps performed

1. Created the folder C:\Shares\MusicLibrary and used the existing LAB\Students group.
2. On the Sharing tab, shared the folder as MusicLibrary (network path \ \SERVER01\MusicLibrary).
3. Under Advanced Sharing > Permissions, the share permission is limited to the Students group.
4. On the Security tab (NTFS), gave Students the Modify permission (read, write and list), applied to this folder, subfolders and files. Administrators keep Full control and no other group has access.

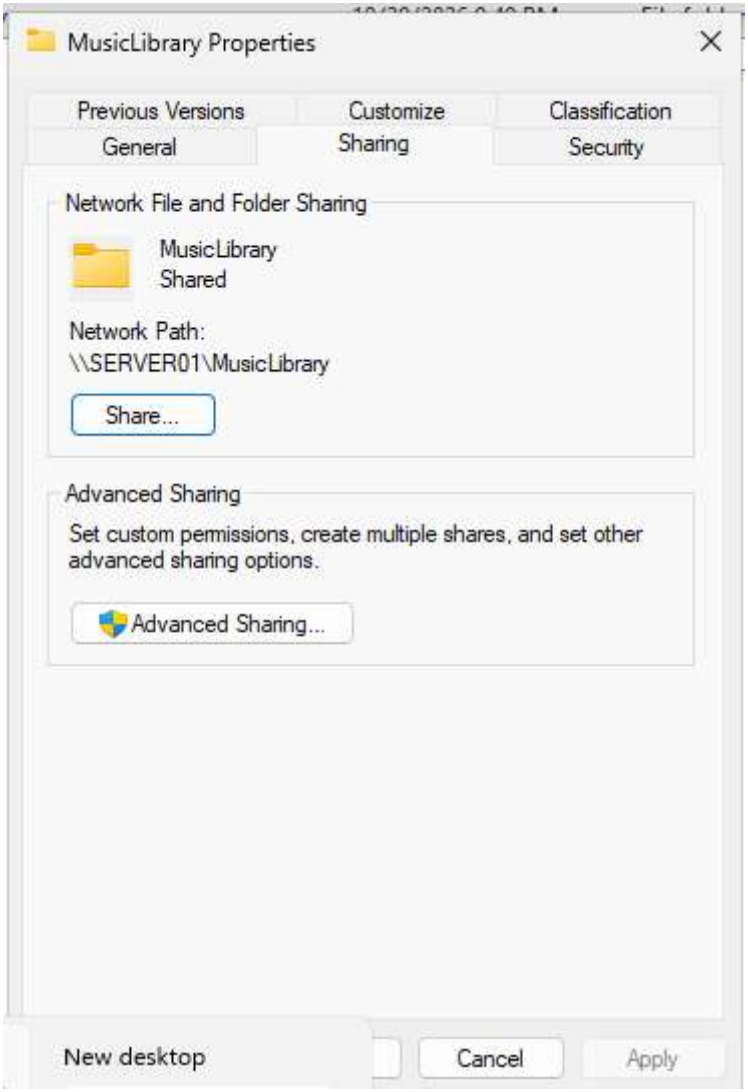


Figure 7: MusicLibrary shared at \\SERVER01\MusicLibrary

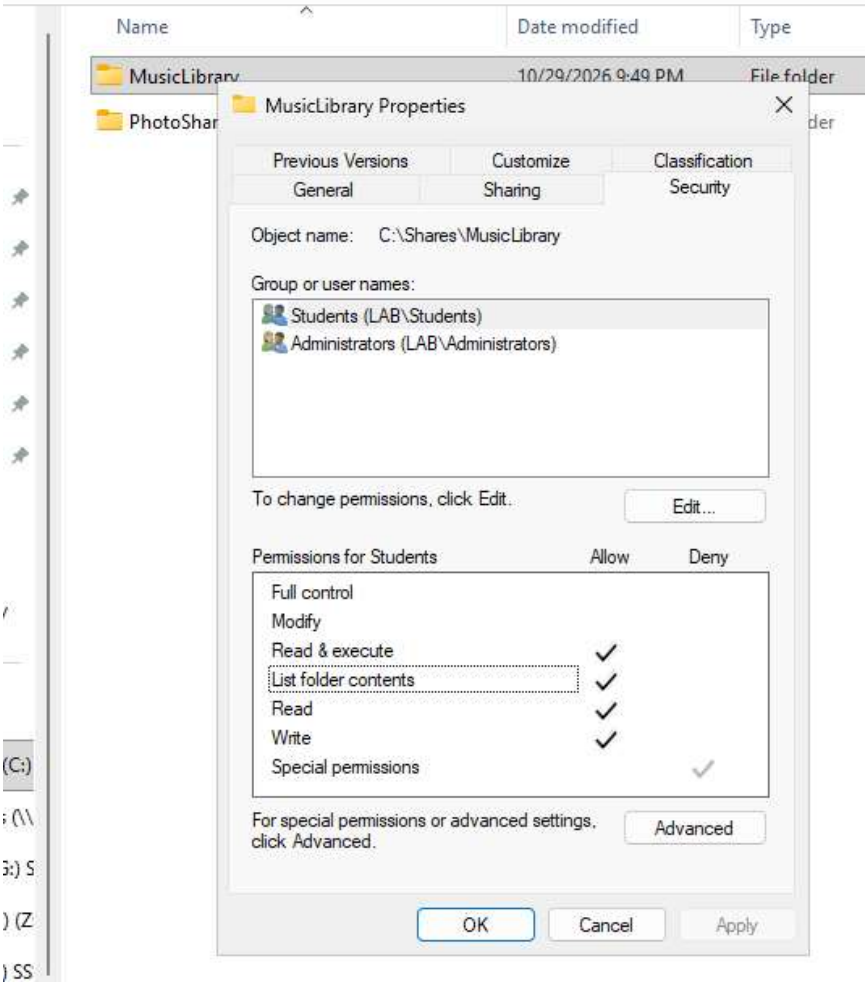


Figure 8: Security tab, Students group with Read and Write permissions

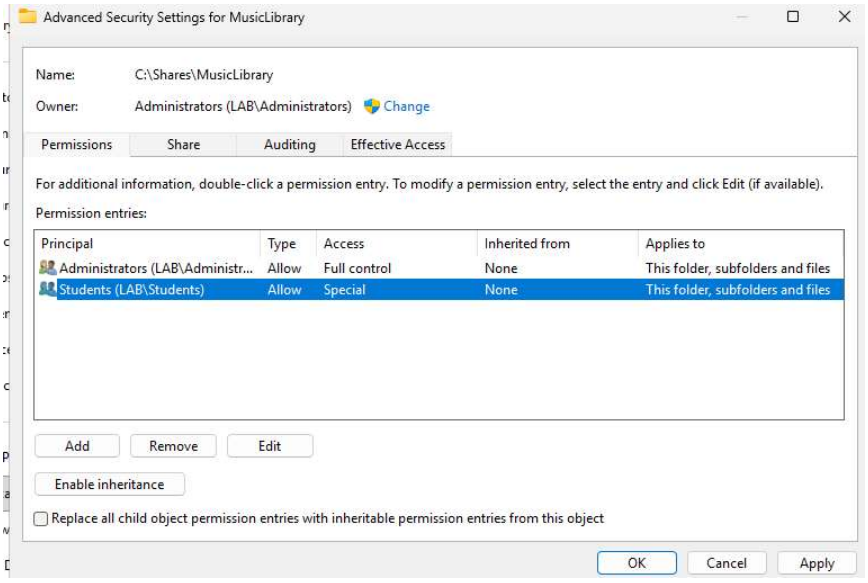


Figure 9: Advanced Security Settings, only Administrators and Students listed

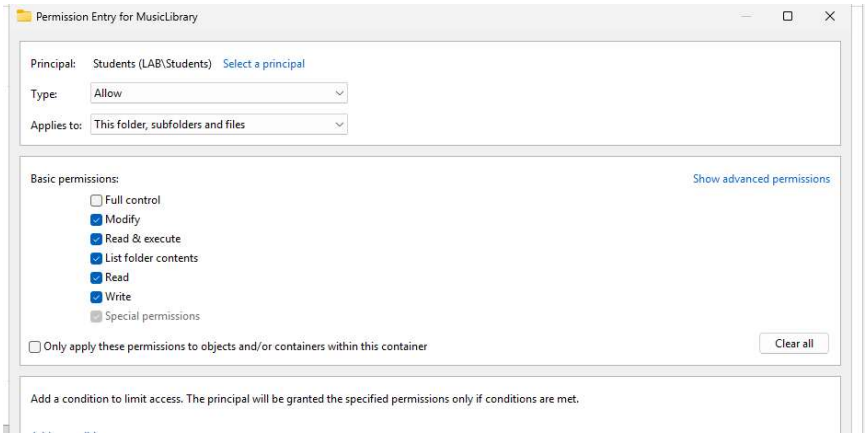


Figure 10: Permission entry, Students with Modify, applies to folder, subfolders and files

Task 4: Install the Print Server role, add a virtual printer (Microsoft XPS Document Writer), and share it as 'CampusPrinter' so network users can print to it.

Steps performed

1. Installed the Print and Document Services role with the Print Server role service.
2. In Devices and Printers, chose Add a printer > The printer I want isn't listed > Add a local printer with manual settings.
3. The Microsoft XPS Document Writer v4 entry was not listed in the driver list. I used the Microsoft XPS Class Driver instead, which is the driver Windows uses for XPS printing.
4. Named the printer CampusPrinter, ticked Share this printer and set the share name to CampusPrinter.

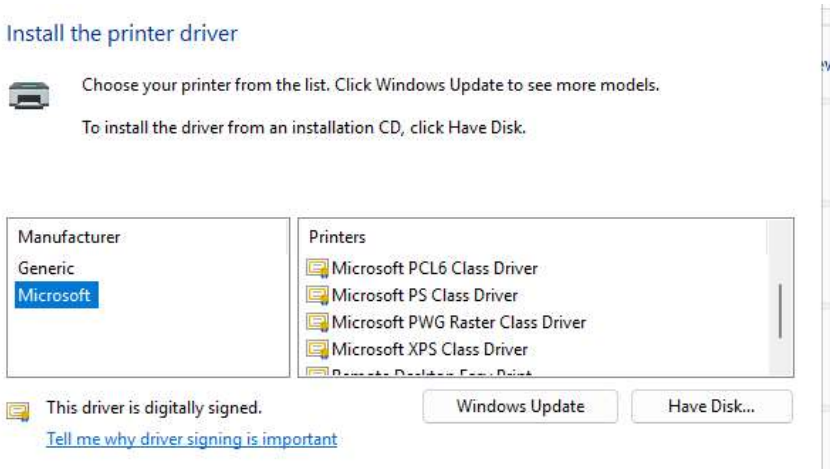


Figure 11: Driver list, Microsoft XPS Class Driver selected

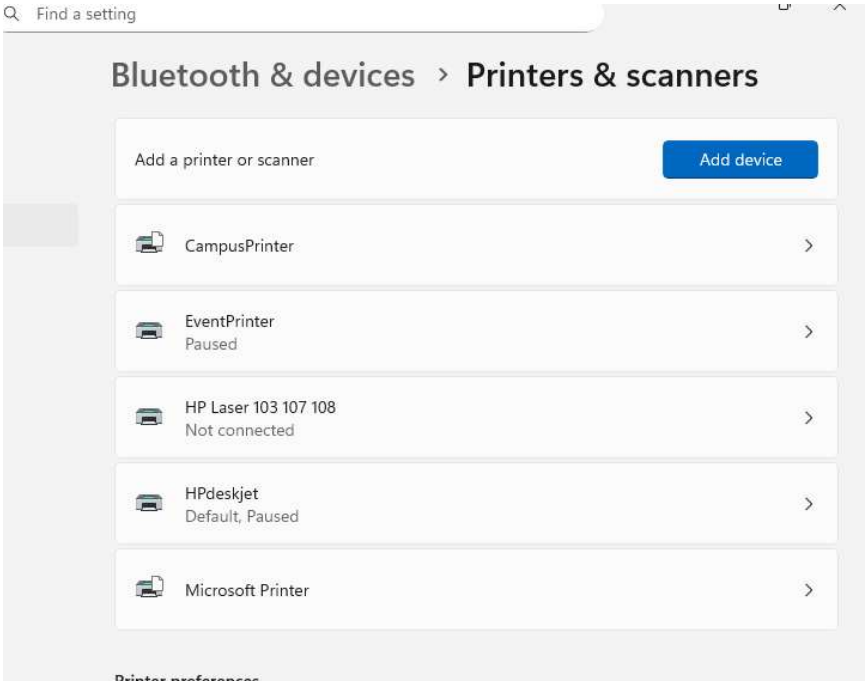


Figure 12: CampusPrinter added in Printers & scanners (not paused)

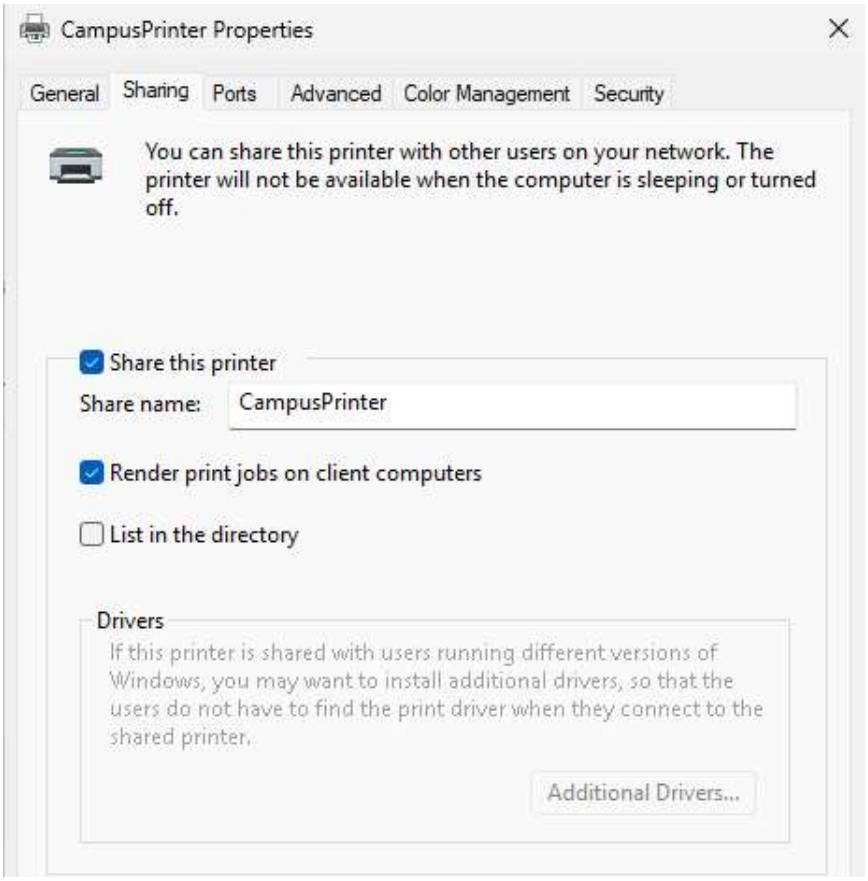


Figure 13: CampusPrinter Sharing tab, Share this printer ticked

Network users connect with \\SERVER01\CampusPrinter and can print to it like any other printer.

Task 5: A client PC is not receiving an IP address from your DHCP server. List 3 possible causes and describe how you would test and fix each one using Windows Server tools.

First check on the client. Run ipconfig /all. An address starting with 169.254.x.x (APIPA) means the client sent a DHCP request but received no reply.

Possible cause	How I would test it	How I would fix it	Tool used
1. DHCP service is stopped, the server is not authorized, or the scope is inactive	Open services.msc and check that the DHCP Server service is Running. In the DHCP console, look for a red down arrow on the server or scope. Run: Get-Service DHCPServer	Start the DHCP Server service. Right-click the server and choose Authorize (if domain joined). Right-click the scope and choose Activate.	Services console, DHCP console, PowerShell
2. The scope has no free addresses left	The scope 192.168.1.100 to .120 holds only 21 addresses. In DHCP, open Address Leases and check how many are in use. Run: Get-DhcpServerv4ScopeStatistics -ScopeId 192.168.1.0 and check Free.	Delete old or expired leases. Extend the address pool range or use a shorter lease time. Remove an exclusion or reservation that is blocking the address.	DHCP console (Address Leases, Address Pool), PowerShell
3. Connectivity or client configuration problem (bad cable or switch port, wrong VLAN, firewall, or the client has a static IP)	On the client, run ipconfig /all and check that DHCP Enabled is Yes. Check the cable, link light and switch port. Ping the server (ping 192.168.1.10). On the server, check Event Viewer and the Windows Firewall rules for DHCP (UDP 67 and 68).	Set the adapter to Obtain an IP address automatically. Fix the cable, port or VLAN. Enable the DHCP firewall rule. Run ipconfig /release then ipconfig /renew.	ipconfig, ping, Event Viewer, Windows Defender Firewall

Verify the fix

```
ipconfig /release
ipconfig /renew
ipconfig /all
```

The client should now show an address between 192.168.1.100 and 192.168.1.120, the server 192.168.1.10 as the DHCP server, and a lease of 8 hours.